

Eg. $f_n \leq \left(\frac{5}{3}\right)^n$

Base case: $n=0$

$$f_0 = 0, \quad \left(\frac{5}{3}\right)^0 = 1$$

$$f_0 \leq \left(\frac{5}{3}\right)^0$$

Inductive hypothesis:

$$f_k \leq \left(\frac{5}{3}\right)^k$$

Inductive step.

To show: $f_{k+1} \leq \left(\frac{5}{3}\right)^{k+1}$
 The hypothesis is true for
 $n = 0, 1, 2, \dots, k-1, k$

$$f_{k+1} = f_k + f_{k-1}$$

$$\leq \left(\frac{5}{3}\right)^k + \left(\frac{5}{3}\right)^{k-1} = \left(\frac{5}{3}\right)^{k-1} \left(\frac{8}{3}\right)$$

$$= \left(\frac{5}{3}\right)^{k-1} \left(\frac{25}{9}\right) \leq \left(\frac{5}{3}\right)^{k-1} \left(\frac{25}{9}\right)$$

$$= \left(\frac{5}{3}\right)^{k-1} \left(\frac{5}{3}\right)^2$$

$$= \left(\frac{5}{3}\right)^{k+1} \quad \square$$

Lucas series: (Lucas no.)

① $L_0 = 2, L_1 = 1.$

② $L_n = L_{n-1} + L_{n-2}$ for $n \in \mathbb{Z}^+, n \geq 2.$

$\forall n, n \in \mathbb{Z}^+, L_n = F_n + F_{n+1}$

Proof: $n=1, n=2.$

Base: $L_1 = 1$

$F_0 + F_2 = 1.$

$L_2 = 3$

$F_1 + F_3 = 3.$

Inductive hypothesis:

$L_k = F_k + F_{k+1}$

$n=1, 2, \dots, k.$

Inductive step:

To show: $L_{k+1} = F_k + F_{k+2}.$

~~$L_{k+1} = L_k + L_{k-1}$~~

~~$= F_{k-2} + F_{k-1} + F_{k-1} + F_k$~~

$L_{k+1} = L_k + L_{k-1}$

$= F_{k-1} + F_{k+1} + F_{k-2} + F_k = F_k + F_{k+2} \quad \square$

The division of integers: certain properties
of primes numbers.

Defⁿ If $a, b \in \mathbb{Z}$ and $b \neq 0$, we say
 b divides a & write it as $b|a$
if there is an integer n such that
 $a = bn$.

b is divisor of a & a is multiple of
 b .

Theorem 1

$[a|b) \wedge (a|c)] \Rightarrow a|(bx+cy)$ for all $x, y \in \mathbb{Z}$
 $bx+cy$ is a linear combination of b & c .

Proof: $a|b$ & $a|c \Rightarrow$ ~~$b = an$~~ $b = an$
 ~~$c = am$~~ $c = am$.

$$bx+cy = (an)x + (am)y = a[nx+my].$$
$$\therefore a|a[nx+my] \quad \square$$

E.g. 1.

$$6x + 9y + 15z = 107$$

Is there any $x, y, z \in \mathbb{Z}$
s.t. the above equation is true.

$$\begin{array}{ccc} 6x & + & 9y & + & 15z & = & 107 \\ \downarrow & & \downarrow & & \downarrow & & \\ \text{multiple of } 3 & & \text{multiple of } 3 & & \text{multiple of } 3 & & \end{array}$$

~~not true.~~
multiple of 3

E.g. 2. $a, b \in \mathbb{Z}$

$2a + 3b$ is a multiple of 17.

Show that 17 divides $9a + 5b$.

$$\rightarrow 17 \mid (2a + 3b)$$

$$\rightarrow 17 \mid (17a + 17b) \quad \text{--- (1)}$$

$$17 \mid (2a + 3b) \Rightarrow 17 \mid (-4)(2a + 3b) \quad \text{--- (2)}$$

From (1) & (2)

$$17 \mid (17a + 17b) + (-4)(2a + 3b)$$
$$\Rightarrow 17 \mid (9a + 5b) \quad \square$$

Lemma 1.

If $n \in \mathbb{Z}^+$ and n is composite then there is prime p s.t. $p|n$.

Proof:

Let there be a set S which has. All composite numbers that do not have prime divisor(s)

We want to show: $S = \emptyset$

By way of contradiction we assume that $S \neq \emptyset$.

By WOP, S must have a least element say m .

Now since m is composite we can write $m = m_1 m_2$ s.t. $1 < m_1 < m$ & $1 < m_2 < m$.

$m_1 \notin S$.

m_1 is either prime or prime divisible as it is not part of S . Then so does m .

If m_1 has a prime divisor p then so does m .
 $\Rightarrow$ contradiction $\Rightarrow S = \emptyset$.

Ther

Theorem 1. (~~Due~~ Due to Euclid).

There are infinitely many primes.

Proof:

By way of contradiction.

Say ~~there~~ there are a finite no. of primes

$$p_1, p_2, \dots, p_k.$$

Let us define $B = p_1 p_2 p_3 \dots p_k + 1$.

B must be composite; since any $p_j, 1 \leq j \leq k$ is less than B .

By Lemma 1 we can say that there is some prime p_s s.t. $p_s \mid B$.

$p_s \mid p_1 \dots p_k$ & $p_s \mid B$
 $\Rightarrow$ ~~$p_s \mid p_1 \dots p_k$~~ $\rightarrow p_s \mid 1 \Rightarrow$ contradiction

$\Rightarrow$ There are infinitely many primes $\square$.

Theorem 2.

The Division Algorithm.

If $a, b \in \mathbb{Z}$ with $b > 0$, then there
exists unique $q, r \in \mathbb{Z}$ with
 $a = qb + r$, $0 \leq r < b$.

Proof:

Existence:

Consider the set $S = \{a - bk \mid k \in \mathbb{Z} \text{ and } a - bk \geq 0\}$.

Assert: S is non-empty.

If $a \geq 0$, we take $k = 0$

then $a - b \cdot 0 = a \geq 0 \therefore a \in S$.

If $a < 0$, take $k = a$.

$$a - ba = a(1 - b)$$

$$(1 - b) \leq 0$$

$$\therefore a(1 - b) \geq 0 \in S.$$

$S \neq \emptyset$.

Assert: S has a least element.

S is a non-empty set of integers $\Rightarrow$ by WOP it should have a least element; call this element r .

$r = a - bq$ for some integer q .

$$\Rightarrow a = bq + r, \quad r \geq 0$$

Assert: $r < b$.

By way of contradiction $r \geq b$.

Consider $r' = r - b$.

$$= a - bq - b$$

$$= a - b(q+1).$$

Since $r \geq b > 0$

$$r' = r - b > 0.$$

$$= a - b(q+1) \geq 0$$

$$\Rightarrow \underline{r' \in S}.$$

But $r' < r \Rightarrow$
least element of S .

~~r~~ is not the
contradiction.

$$\therefore r < b. \quad \square.$$

der are

$\square$.

~~Uniqueness~~ Uniqueness

By way of contradiction:

Let $q_1, q_2, r_1, r_2 \in \mathbb{Z}$
The q & r are not unique

$$a = q_1 b + r_1$$

$$0 \leq r_1 < b$$

AND

$$a = q_2 b + r_2$$

$$0 \leq r_2 < b$$

$$q_1 b + r_1 = q_2 b + r_2$$

$$\Rightarrow \underline{b | q_1 - q_2} = |r_1 - r_2| < b$$

$$b | q_1 - q_2 < b \text{ cannot happen}$$

unless $q_1 = q_2$ & so $r_1 = r_2$

Thus the quotient & the remainder are
unique.

□.

E.g.

$\Rightarrow$ If $n \in \mathbb{Z}^+$ and n is composite then there exists a prime p s.t. $p|n$ & $p \leq \sqrt{n}$

Defⁿ 2

Since n is composite we can write
$$n = n_1 n_2 \quad , \quad \text{s.t. } 1 < n_1 < n \quad \& \quad 1 < n_2 < n.$$

Defⁿ

We claim that one of them would be $< \sqrt{n}$.

If no. then by way of contradiction
 $n_1 > \sqrt{n} \quad , \quad n_2 > \sqrt{n} \Rightarrow n_1 n_2 > n.$

contradiction
 $\therefore$ either n_1 or n_2 is less than $\sqrt{n}$.

WLOG, $n_1 < \sqrt{n}$.

If n_1 is prime: then we ~~are~~ are done.

If n_1 is composite $\Rightarrow$ ~~not~~ use

Lemma 1 $\Rightarrow$ there should be some

prime p s.t. $p|n_1$
 $\Rightarrow p|n$ & $p \leq \sqrt{n}$. $\square$

Properties of common divisors & greatest common divisor in specific.

Defⁿ 2 For $a, b \in \mathbb{Z}$, a positive integer c is said to be a common divisor of a, b if $c|a$ & $c|b$.

Defⁿ 3 Let $a, b \in \mathbb{Z}$ where either $a \neq 0$ or $b \neq 0$. Then $c \in \mathbb{Z}^+$ is called the gcd of a, b if $c|a, c|b$ (c is a common divisor of a, b)

(a) $c|a, c|b$

(b) for any common divisor d of a, b we have $\frac{d}{c}$

Theorem: For any $a, b \in \mathbb{Z}^+$, there exists a unique $c \in \mathbb{Z}^+$ that is the gcd of a, b .

Existence.

Define $S = \{ax + by \mid x, y \in \mathbb{Z}, ax + by > 0\}$.

Step 1: S is non-empty:

$$x=1, y=0 \Rightarrow a \cdot 1 + b \cdot 0 = a > 0 \in S.$$

Step 2: S has a least element by WOP.

Let this be c

$$c = ax_0 + by_0$$

→ Bezout's Identity

Step 3: To show $c|a$ & $c|b$.

By div. algorithm

$$a = qc + r \quad 0 \leq r < c.$$

$$r = a - qc.$$

$$= a - q(ax_0 + by_0).$$

$$= a(1 - qx_0) + b(-qy_0).$$

r is of the form $ax + by$

If $r > 0$ then $r \in S$.

r becomes the least element & not c .
• $\therefore r \neq 0$.
Also $r \neq 0$ (by division algo).

$\therefore r = 0$.

$$a = qc \Rightarrow c/a.$$

By similar argument
Uniqueness.

~~c/b~~ $\square$.

By way of contradiction:

Let c_1 & c_2 be two gcds of a, b .

$$\Rightarrow c_1/c_2 \text{ \& } c_2/c_1$$

$$\Rightarrow c_1 = c_2$$

$\square$.

Euclidean Algorithm to find gcd.

Algorithm:

- ① At every step divide the previous divisor with the previous remainder.
- ② Continue ① until we arrive at remainder 0.
- ③ The last ~~non-zero~~ non-zero remainder is the gcd.

$$a = q_1 b + r_1 \quad 0 < r_1 < b.$$

$$b = q_2 r_1 + r_2 \quad 0 < r_2 < r_1$$

$$r_1 = q_3 r_2 + r_3 \quad 0 < r_3 < r_2$$

$$\vdots$$

$$r_i = q_{i+2} r_{i+1} + r_{i+2} \quad 0 < r_{i+2} < r_{i+1}$$

$$r_{k-2} = q_k r_{k-1} + r_k$$

$$r_{k-1} = q_{k+1} r_k + 0$$

$$r_k = \text{gcd}$$

Claim: The algorithm should terminate.

$$b > r_1 > r_2 \dots \geq 0$$

is a strictly decreasing series so by WOP
it must reach 0.

Claim: The algorithm is correct.

$$a = qb + r$$

To show: $\gcd(a, b) = \gcd(b, r)$
 $\gcd(a, b) \mid \gcd(b, r)$

$$a = qb + r$$

$$\Rightarrow r = a - qb$$

Let d be a common divisor of
 $a, b \Rightarrow d \mid a$ & $d \mid b$.

$$\Rightarrow d \mid r.$$

So d is a common divisor of
 b & r .

$$\& \gcd(a, b) > \gcd(b, r).$$

Let $d \mid b$ & $d \mid r$.

Since $a = bq + r$

$$\Rightarrow d \mid a.$$

Thus $d \mid a$ & $d \mid b$.

Hence $\{d: d \mid b, d \mid r\} = \{d: d \mid a, d \mid b\}$
The two sets have the same
max element $\therefore \gcd(a, b) = \gcd(b, r)$